LAB NOC 11

Cisco 888 ISR — DHCP Configuration & Troubleshooting

Real Hardware | ES / EN | NOC Portfolio

INFORMACION DEL LABORATORIO / LAB INFORMATION

Hardware	Cisco Catalyst 2960 + Cisco 888 ISR	Cisco Catalyst 2960 + Cisco 888 ISR
Ticket ID	NOC-LAB-11	NOC-LAB-11
Categoria / Category	DHCP — Configuracion y troubleshooting	DHCP — Configuration and troubleshooting
Nivel / Level	CCNA — DHCP / IP Services	CCNA — DHCP / IP Services
Topologia / Topology	PC → Switch Fa0/7 → Switch Fa0/16 → Router 888	PC → Switch Fa0/7 → Switch Fa0/16 → Router 888
Pool DHCP	192.168.10.11 – 192.168.10.254 (excluidas .1–.10)	192.168.10.11 – 192.168.10.254 (excluded .1–.10)
Duracion / Duration	Aprox. 35 minutos	Approx. 35 minutes
Relevancia NOC	Diagnostico de falla DHCP, APIPA, renovacion de lease	DHCP failure diagnosis, APIPA, lease renewal

OBJETIVO / OBJECTIVE

OBJETIVO	OBJECTIVE
Configurar un servidor DHCP en el Cisco 888 ISR para asignar direcciones IP automaticamente a clientes de la red 192.168.10.0/24, simular la falla del servicio DHCP, diagnosticar el sintoma APIPA en la PC, y restaurar el servicio documentando el incidente en formato NOC.	Configure a DHCP server on the Cisco 888 ISR to automatically assign IP addresses to clients on the 192.168.10.0/24 network, simulate DHCP service failure, diagnose the APIPA symptom on the PC, and restore the service documenting the incident in NOC format.

INCIDENT TICKET — NOC-LAB-11

CAMPO / FIELD	DETALLE / DETAIL
Ticket ID	NOC-LAB-11
Title / Titulo	Usuario sin conectividad de red — PC sin IP / User without network connectivity — PC has no IP
Priority / Prioridad	Medium — usuario productivo sin acceso a red / productive user without network access
Symptom	PC muestra IP 169.254.X.X (APIPA) — no obtiene IP del servidor DHCP / PC shows IP 169.254.X.X (APIPA) — not receiving IP from DHCP server
Root Cause	Servicio DHCP deshabilitado en el router (no service dhcp) / DHCP

	service disabled on router (no service dhcp)
Diagnosis Command	show running-config | include dhcp → confirma 'no service dhcp' activo / confirms 'no service dhcp' active
Resolution	service dhcp en el router + ipconfig /release + ipconfig /renew en la PC / service dhcp on router + ipconfig /release + ipconfig /renew on PC
Validation	ipconfig → IP 192.168.10.X + show ip dhcp binding → MAC de PC con IP asignada
Status	CERRADO — DHCP restaurado, PC con IP correcta / CLOSED — DHCP restored, PC with correct IP

CONFIGURACION DEL POOL DHCP / DHCP POOL CONFIGURATION

PARAMETRO / PARAMETER	VALOR / VALUE	DESCRIPCION / DESCRIPTION
Pool Name	`LAN`	Nombre del pool DHCP / DHCP pool name
Network	`192.168.10.0 255.255.255.0`	Red asignada por el pool / Network assigned by pool
Default Router	`192.168.10.1`	Gateway entregado a los clientes / Gateway delivered to clients
DNS Server	`8.8.8.8`	Servidor DNS de Google entregado a clientes / Google DNS server delivered to clients
Excluded Range	`192.168.10.1 – 192.168.10.10`	Reservadas para equipos de red (router, switches) / Reserved for network devices
Assignable Range	`192.168.10.11 – 192.168.10.254`	Rango disponible para asignacion automatica / Range available for automatic assignment
SVI / Gateway	`interface vlan 1 → 192.168.10.1 /24`	La SVI del router actua como gateway del pool / Router SVI acts as pool gateway

PROCEDIMIENTO — 5 ETAPAS / PROCEDURE — 5 STAGES

#	ESPANOL	ENGLISH
1	ETAPA 1 — Configurar DHCP en el router: ip dhcp excluded-address 192.168.10.1 192.168.10.10	STAGE 1 — Configure DHCP on router: ip dhcp excluded-address 192.168.10.1 192.168.10.10
2	ip dhcp pool LAN → network 192.168.10.0 /24 → default-router 192.168.10.1 → dns-server 8.8.8.8	ip dhcp pool LAN → network 192.168.10.0 /24 → default-router 192.168.10.1 → dns-server 8.8.8.8
3	Configurar SVI: interface vlan 1 → ip address 192.168.10.1 /24 → no shutdown → write memory	Configure SVI: interface vlan 1 → ip address 192.168.10.1 /24 → no shutdown → write memory
4	ETAPA 2 — Configurar PC en modo DHCP automatico (Panel de control → Adaptador → Obtener IP automaticamente)	STAGE 2 — Configure PC in automatic DHCP mode (Control Panel → Adapter → Obtain IP automatically)
5	Verificar: ipconfig → PC recibe 192.168.10.X,	Verify: ipconfig → PC receives 192.168.10.X,

	mask /24, gateway 192.168.10.1	mask /24, gateway 192.168.10.1
6	Verificar en router: show ip dhcp binding → MAC de PC con IP asignada	Verify on router: show ip dhcp binding → PC MAC with assigned IP
7	ETAPA 3 — Simular falla: conf t → no service dhcp → end	STAGE 3 — Simulate failure: conf t → no service dhcp → end
8	Liberar IP de la PC: ipconfig /release → ipconfig /renew → observar IP APIPA 169.254.X.X	Release PC IP: ipconfig /release → ipconfig /renew → observe APIPA IP 169.254.X.X
9	ETAPA 4 — Diagnostico: show running-config | include dhcp → confirmar 'no service dhcp'	STAGE 4 — Diagnosis: show running-config | include dhcp → confirm 'no service dhcp'
10	ETAPA 5 — Resolucion: conf t → service dhcp → end	STAGE 5 — Resolution: conf t → service dhcp → end
11	En PC: ipconfig /release → ipconfig /renew → confirmar IP 192.168.10.X	On PC: ipconfig /release → ipconfig /renew → confirm IP 192.168.10.X
12	Validar: show ip dhcp binding → MAC presente con IP. Ticket NOC-LAB-11 cerrado.	Validate: show ip dhcp binding → MAC present with IP. Ticket NOC-LAB-11 closed.

DHCP Troubleshooting

Objetivo del laboratorio

Verificar que una PC obtenga automáticamente una dirección IP desde el router mediante **DHCP** y simular una falla del servicio para diagnosticar el problema.

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1. Topología

PC
|
Fa0/7
|
Switch Cisco 2960
|
Fa0/16
|
Router Cisco 888

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2. Escenario

Un usuario reporta que **su computadora no tiene conexión a la red**.
El técnico NOC debe verificar si el problema está relacionado con **DHCP**.

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3. Configuración DHCP en el Router

Ingresar al router:

enable
configure terminal

Excluir direcciones que no serán entregadas por DHCP:

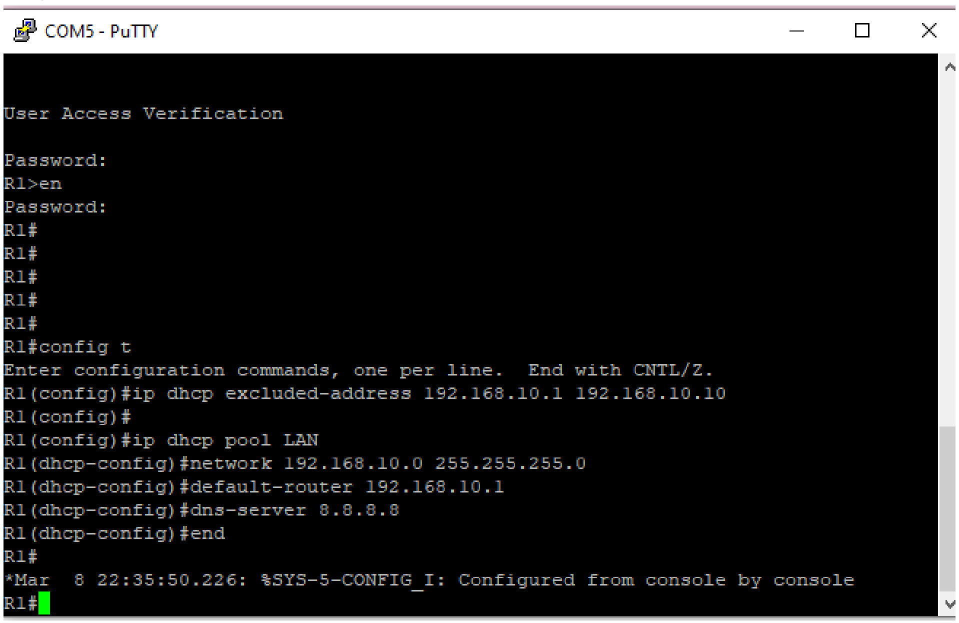
ip dhcp excluded-address 192.168.10.1 192.168.10.10

Crear el pool DHCP:

ip dhcp pool LAN
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8

Salir:

End



4. Configurar IP en la interfaz del Router

La interfaz conectada al switch debe tener IP.

Ejemplo:

configure terminal
interface vlan 1
ip address 192.168.10.1 255.255.255.0
no shutdown
end

```
R1#
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface vlan 1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#end
R1#
*Mar  8 22:47:43.851: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

5. Verificar el servicio DHCP

Comando para ver direcciones asignadas:

show ip dhcp binding

Comando para ver el pool DHCP:

show ip dhcp pool

```
R1#
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
R1#show ip dhcp pool

Pool LAN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
 Pending event                  : none
 1 subnet is currently in the pool :
 Current index        IP address range                    Leased addresses
 192.168.10.1         192.168.10.1     - 192.168.10.254    0
R1#
```

Comando para ver estadísticas DHCP:

show ip dhcp server statistics

COM5 - PuTTY

```
R1#show ip dhcp server statistics
Memory usage         23944
Address pools        1
Database agents      0
Automatic bindings   0
Manual bindings      0
Expired bindings     0
Malformed messages   0
Secure arp entries   0

Message              Received
BOOTREQUEST          0
DHCPDISCOVER         0
DHCPREQUEST          0
DHCPDECLINE          0
DHCPRELEASE          0
DHCPINFORM           0

Message              Sent
BOOTREPLY            0
DHCPOFFER            0
DHCPACK              0
DHCPNAK              0
R1#
```

```
COM5 - PuTTY
R1#
R1#
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
BRI0                       unassigned      YES NVRAM  administratively down down

BRI0:1                     unassigned      YES unset  administratively down down

BRI0:2                     unassigned      YES unset  administratively down down

FastEthernet0              unassigned      YES unset  up                    up

FastEthernet1              unassigned      YES unset  up                    down

FastEthernet2              unassigned      YES unset  up                    down

FastEthernet3              unassigned      YES unset  up                    down

Vlan1                      192.168.10.1    YES manual up                    up

Vlan10                     unassigned      YES unset  down                  down

R1#
```

6. Configuración de la PC

En Windows:

Panel de control
Red e Internet
Centro de redes
Cambiar configuración del adaptador

Configurar:

Obtener una dirección IP automáticamente

7. Verificación en la PC

En la terminal de Windows:

ipconfig

Resultado esperado:

IPv4 Address : 192.168.10.X
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.10.1

```
Símbolo del sistema
Adaptador de Ethernet Ethernet:

   Sufijo DNS específico para la conexión. . :
   Vínculo: dirección IPv6 local. . . : fe80::a6db:9b98:bb65:9d10%7
   Dirección IPv4. . . . . . . . . . . . . . : 192.168.10.10
   Máscara de subred . . . . . . . . . . . . : 255.255.255.0
   Puerta de enlace predeterminada . . . . . : 192.168.10.1
```

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8. Simulación de falla DHCP

Para simular un incidente de red:

En el router:

configure terminal
no service dhcp
end

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9. Síntoma observado

La PC obtiene una IP automática **APIPA**:

169.254.X.X

Esto indica que **no hay servidor DHCP disponible en la red**.

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10. Diagnóstico NOC

El técnico verifica:

1 El router está encendido
2 La interfaz está activa
3 El servicio DHCP está habilitado

Comando para verificar:

show running-config | include dhcp

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11. Solución

Reactivar DHCP:

configure terminal
service dhcp
end

Renovar IP en la PC:

ipconfig /release
ipconfig /renew

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12. Validación final

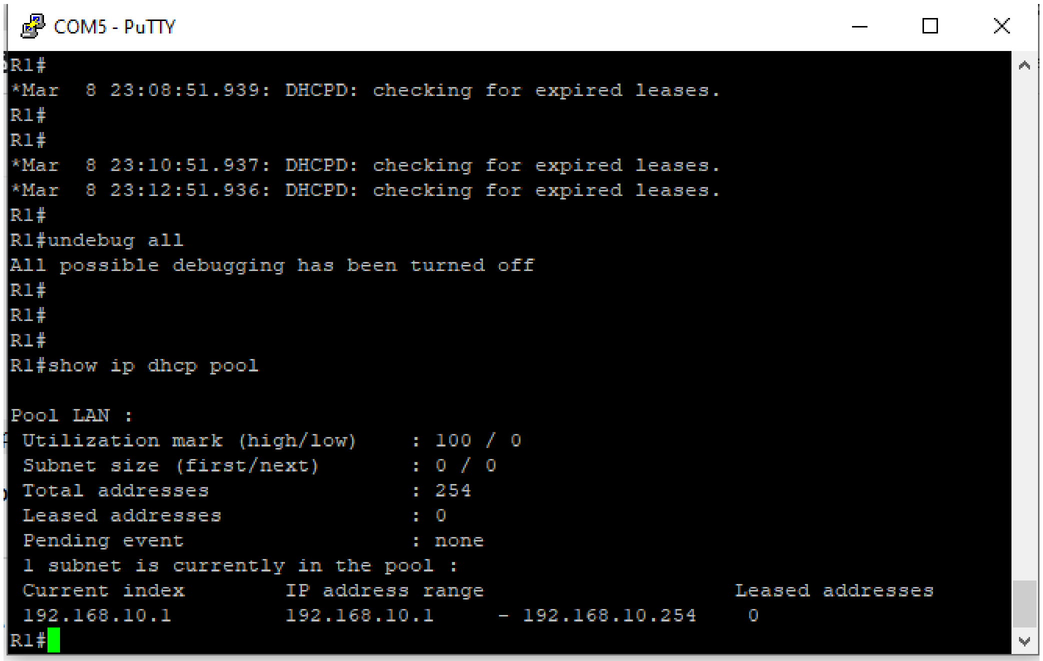
Comprobar que la PC obtiene una IP correcta:

192.168.10.X
Gateway: 192.168.10.1

Verificar en el router:

show ip dhcp binding

Debe aparecer la MAC de la PC con su IP asignada.



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13. Conclusión del incidente NOC
Incidente:
Usuario sin conectividad de red.
Causa:
Servicio DHCP deshabilitado en el router.
Resolución:
Se habilitó nuevamente el servicio DHCP y se renovó la dirección IP del cliente.
Estado final:
Servicio restaurado.

CONFIGURACION DHCP PASO A PASO / STEP-BY-STEP DHCP CONFIGURATION

#	COMANDO / COMMAND	ACCION (ES)	ACTION (EN)
1	`ip dhcp excluded-address 192.168.10.1 192.168.10.10`	Excluir IPs reservadas para equipos de red	Exclude IPs reserved for network devices
2	`ip dhcp pool LAN`	Crear pool DHCP llamado LAN	Create DHCP pool named LAN
3	`network 192.168.10.0 255.255.255.0`	Definir la red que el pool va a servir	Define the network the pool will serve
4	`default-router 192.168.10.1`	Entregar el gateway a los clientes DHCP	Deliver gateway to DHCP clients
5	`dns-server 8.8.8.8`	Entregar DNS a los clientes	Deliver DNS to clients
6	`exit → end`	Salir del pool y del modo configuracion	Exit pool and configuration mode
7	`interface vlan 1 → ip address 192.168.10.1 /24 → no shutdown`	Configurar SVI del router como gateway	Configure router SVI as gateway
8	`write memory`	Persistir toda la configuracion DHCP	Persist all DHCP configuration

ESCENARIO DE FALLA — DHCP DESHABILITADO / FAILURE SCENARIO

FASE / PHASE	ESPANOL	ENGLISH
Baseline	DHCP activo. PC obtiene 192.168.10.X automaticamente. show ip dhcp binding muestra MAC + IP. / DHCP active. PC automatically gets 192.168.10.X. show ip dhcp binding shows MAC + IP.	DHCP active. PC automatically receives 192.168.10.X. show ip dhcp binding shows MAC + IP.
Falla inducida / Induced failure	conf t → no service dhcp → end (servicio DHCP deshabilitado en el router) / conf t → no service dhcp → end (DHCP service disabled on router)	conf t → no service dhcp → end (DHCP service disabled on router)
Sintoma / Symptom	PC obtiene IP APIPA: 169.254.X.X — indica ausencia de servidor DHCP en la red. Sin conectividad. / PC gets APIPA IP: 169.254.X.X — indicates no DHCP server on network. No connectivity.	PC gets APIPA IP: 169.254.X.X — indicates no DHCP server on network. No connectivity.
Diagnostico NOC / Diagnosis	show running-config | include dhcp → muestra 'no service dhcp'. Servicio deshabilitado confirmado. / show running-config | include dhcp → shows 'no service dhcp'. Service disabled confirmed.	show running-config | include dhcp → shows 'no service dhcp'. Service disabled confirmed.
Resolucion / Resolution	conf t → service dhcp → end. En la PC: ipconfig /release → ipconfig /renew. / conf t → service dhcp → end. On PC: ipconfig /release → ipconfig /renew.	conf t → service dhcp → end. On PC: ipconfig /release → ipconfig /renew.
Validacion /	ipconfig → IP 192.168.10.X (rango	ipconfig → IP 192.168.10.X (correct

Validation	correcto). show ip dhcp binding → MAC de PC con IP asignada. Ticket cerrado. / ipconfig → IP 192.168.10.X (correct range). show ip dhcp binding → PC MAC with assigned IP. Ticket closed.	range). show ip dhcp binding → PC MAC with assigned IP. Ticket closed.

APIPA — DIAGNOSTICO RAPIDO / QUICK DIAGNOSIS

IP / CONCEPTO	NOMBRE / NAME	SIGNIFICADO / MEANING
`169.254.X.X`	**APIPA (Automatic Private IP Addressing)**	Windows asigna automaticamente cuando no hay servidor DHCP disponible en la red / Windows automatically assigns when no DHCP server is available on the network
`Rango / Range`	**169.254.0.0 / 16**	Rango reservado por IANA para APIPA — no enrutable en Internet / IANA reserved range for APIPA — not routable on Internet
`Diagnostico clave / Key diagnosis`	**PC con 169.254.X.X = sin DHCP**	Verificar: (1) router encendido, (2) interfaz up, (3) service dhcp activo / Check: (1) router on, (2) interface up, (3) service dhcp active
`En mineria / In mining`	**IP de camara, PLC o terminal SCADA en 169.254.X.X**	Dispositivo OT sin DHCP — verificar pool, interfaz SVI y estado del servicio DHCP antes de escalar / OT device without DHCP — check pool, SVI interface and DHCP service status before escalating

REFERENCIA DE COMANDOS / COMMAND REFERENCE

COMANDO / COMMAND	SIGNIFICADO (ES)	MEANING (EN)
`show ip dhcp binding`	Muestra todas las IPs asignadas: MAC, IP, expiracion, tipo	Shows all assigned IPs: MAC, IP, expiration, type
`show ip dhcp pool`	Muestra configuracion del pool: red, gateway, DNS, rango	Shows pool configuration: network, gateway, DNS, range
`show ip dhcp server statistics`	Estadisticas: mensajes DISCOVER/OFFER/REQUEST/ACK enviados	Statistics: DISCOVER/OFFER/REQUEST/ACK messages sent
`show running-config | include dhcp`	Filtro rapido — muestra si 'no service dhcp' esta activo	Quick filter — shows if 'no service dhcp' is active
`service dhcp / no service dhcp`	Habilitar / deshabilitar el servicio DHCP en el router	Enable / disable DHCP service on the router
`ip dhcp excluded-address X Y`	Excluir rango de IPs de la asignacion automatica	Exclude IP range from automatic assignment
`ipconfig /release`	PC Windows — libera la IP DHCP actual	PC Windows — releases current DHCP IP
`ipconfig /renew`	PC Windows — solicita nueva IP al servidor DHCP	PC Windows — requests new IP from DHCP server

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

HALLAZGO / FINDING	ACCION CORRECTIVA / CORRECTIVE ACTION
PC obtiene 169.254.X.X (APIPA) en vez de 192.168.10.X — sintoma clasico de servidor DHCP no disponible. / PC gets 169.254.X.X (APIPA) instead of 192.168.10.X — classic symptom of DHCP server unavailable.	Verificar en orden: (1) router encendido y accesible por consola, (2) show ip interface brief — SVI up/up, (3) show running-config | include dhcp — confirmar que 'no service dhcp' NO aparece. / Verify in order: (1) router on and accessible via console, (2) show ip interface brief — SVI up/up, (3) show running-config | include dhcp — confirm 'no service dhcp' does NOT appear.
La PC obtiene IP DHCP pero esta fuera del rango esperado — por ejemplo obtiene una IP que deberia estar excluida. / PC gets DHCP IP but outside expected range — for example gets an IP that should be excluded.	Verificar con show ip dhcp pool la lista de exclusiones. Si falta el ip dhcp excluded-address, los equipos de red pueden recibir IPs que ya estan asignadas estaticamente al router o switches, generando conflictos. / Verify with show ip dhcp pool the exclusion list. If ip dhcp excluded-address is missing, network devices may receive IPs already statically assigned to routers or switches, causing conflicts.
Despues de service dhcp la PC aun muestra 169.254.X.X aunque el servidor esta activo. / After service dhcp the PC still shows 169.254.X.X even though server is active.	El cliente DHCP de Windows necesita liberar y renovar explicitamente: ipconfig /release seguido de ipconfig /renew. Sin este paso la PC no solicita nueva IP aunque el servidor este disponible. / Windows DHCP client needs explicit release and renew: ipconfig /release followed by ipconfig /renew. Without this step the PC does not request a new IP even if the server is available.

RESULTADO / RESULT

RESULTADO (ES)	RESULT (EN)
DHCP configurado y operativo en Cisco 888. Pool LAN asignando 192.168.10.11-254. Falla simulada (no service dhcp) diagnosticada via APIPA. Servicio restaurado con service dhcp + ipconfig /renew. Ticket NOC-LAB-11 cerrado.	DHCP configured and operational on Cisco 888. LAN pool assigning 192.168.10.11-254. Simulated failure (no service dhcp) diagnosed via APIPA. Service restored with service dhcp + ipconfig /renew. Ticket NOC-LAB-11 closed.

KEY TAKEAWAY

"En NOC, el primer indicador de falla DHCP es una IP 169.254.X.X (APIPA) en el cliente. El diagnostico es inmediato: show running-config | include dhcp en el router. Si aparece 'no service dhcp', un solo comando lo resuelve. El error mas comun post-resolucion es olvidar el ipconfig /renew en la PC — el servidor puede estar activo pero el cliente Windows no solicita nueva IP hasta que se le indica explicitamente."

"In NOC, the first indicator of DHCP failure is a 169.254.X.X (APIPA) IP on the client. Diagnosis is immediate: show running-config | include dhcp on the router. If 'no service dhcp' appears, a single command resolves it. The most common post-resolution error is forgetting ipconfig /renew on the PC — the server may be active but the Windows client does not request a new IP until explicitly told to."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining operation (e.g., Newmont Cerro Negro), DHCP failures are high-frequency tickets — particularly after router reloads, maintenance windows, or configuration changes that accidentally disable the service. A NOC engineer must recognize APIPA (169.254.X.X) as an immediate DHCP failure signature, know the three-command diagnosis sequence (show ip dhcp binding → show ip dhcp pool → show running-config | include dhcp), and understand that OT devices (IP cameras, PLCs, SCADA terminals) configured for DHCP will lose connectivity silently if the service is disabled. For critical OT devices, static IP assignment with DHCP exclusion of those addresses is the recommended practice to avoid accidental IP conflicts.